

Motivation

Reil.App is a mobile service that generates daily meal recommendations aligned with user preferences and macro-nutrient (KBJU) targets, auto-assembles the cart and places an order via delivery partners - no manual counting, no complex diets. *Industry data: ~70% of a user's food-related time goes to planning and shopping; Reil removes that layer.*

Its candidate-generation core is **text-embedding based**. Before committing to an architecture we ran a controlled study on a synthetic dataset designed to mirror Reil.App production signals (KBJU targets, allergens, goal types, archetype diversity, family-group patterns).

Question. Which embedding components actually move item-to-item retrieval quality, and which are expensive ceremony? The answer shaped the production pipeline shown in Figure 1 and - concretely - **motivated promoting the behavioral-vector similarity to the dominant weight** in the current weighted-sum scorer, until enough real interaction data accumulates to train a heavy LTR ranker (LightGBM / two-tower).

Experimental Setup

Dataset — synthetic, mirrors Reil.App signals

- 4,939 LLM-generated dishes · 8 cuisine clusters
- 1,100 users across 10 archetypes (fitness · diabetic · vegetarian · gluten-free · allergy-prone · family-mom · ...)
- Per-user fields reproducing Reil.App profile: **KBJU targets**, allergens, goal-type (lose/keep/gain), price tier
- 195,749 interactions (view / order / rating / favorite) · 60/20/20 split
- Signal validation: calorie-goal alignment, 0 allergen violations in orders/favorites

Models

- Dense encoder: paraphrase-multilingual-mpnet-base-v2 · 768-d
- Cross-encoder: ms-marco-MiniLM-L-6-v2
- LLM reranker: gpt-4o-mini (OpenRouter, JSON-mode)
- Lexical baseline: TF-IDF + TruncatedSVD(256)

Protocol — item-to-item co-preference

- For each test user with ≥ 5 positives (orders / favorites)
- One positive dish q is the **query**; the rest is the **relevant set** R_u
- Retrieve top- K from the full 4,939-dish catalog
- Report $P@K$, $NDCG@K$, $HR@K$, MRR - averaged over users

Absolute $P@10$ is structurally small ($K=10$ vs. few positives) - **MRR and $NDCG$ are the more informative signals.**

Methods

A Parametrized dish-text builder

All text variants are produced by flagged calls to one function `dish_to_rich_text(d, ·)`. Fields: `DISH_NAME`, `DESCRIPTION`, `NUTRITION`, `MACRO_TAGS`, `TAGS`, `INGREDIENTS`, `RECIPE_BRIEF`. Ablated via boolean flags: `include_recipe`, `include_macro_tokens`, `include_ratios`, `include_ingredients`.

B Behavioral user embedding

Instead of encoding a user-profile *text*, aggregate the user's interacted dish vectors directly:

$$\mathbf{e}_u = \frac{\sum_{i \in \mathcal{H}_u} w_i \mathbf{e}_{d_i}}{\left\| \sum_{i \in \mathcal{H}_u} w_i \mathbf{e}_{d_i} \right\|_2}, \quad w_i \in \{w_{\text{order}}, w_{\text{fav}}, w_{\text{rating}}\}$$

Closes the semantic gap between profile-text space and dish-text space.

C Hybrid retrieval - BM25 + Dense via RRF

$$\text{RRF}(d) = \sum_{r \in \{\text{bm25}, \text{dense}\}} \frac{1}{k + \text{rank}_r(d)}, \quad k = 60$$

D Domain fine-tuning - MNRL

In-batch negatives; pairs = two dishes the same train user ordered / favored.

$$\mathcal{L}_{\text{MNRL}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(\mathbf{a}_i, \mathbf{p}_i) / \tau)}{\sum_{j=1}^B \exp(\text{sim}(\mathbf{a}_i, \mathbf{p}_j) / \tau)}$$

27,160 pairs · 4 epochs · batch 64 · Tesla T4.

E LLM reranker with user profile

Two-stage: dense retrieval $\rightarrow$ top- N pool $\rightarrow$ LLM ranks candidates conditioned on `USER_PROFILE` + `QUERY_DISH` with a strict-JSON schema ($N=50$, gpt-4o-mini). Fair-compare against *dense top-pool-only* on the same pool and users.

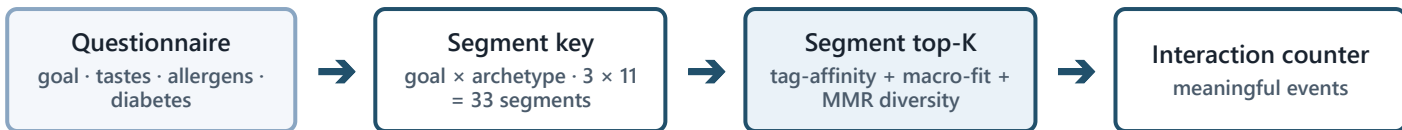
Acknowledgements · Artifacts

Code, notebooks and a synthetic-data generator are available in the project repository. Each experiment is self-contained and reproducible on a 2x T4 GPU (Kaggle / Colab).

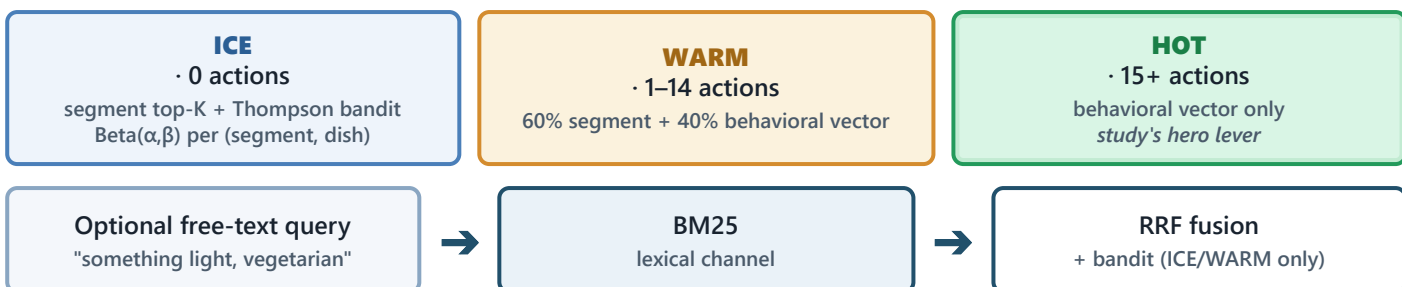
Base encoder & cross-encoder from sentence-transformers · FAISS for ANN · BM25 via rank_bm25 · gpt-4o-mini via OpenRouter.

Reil.App Production Architecture

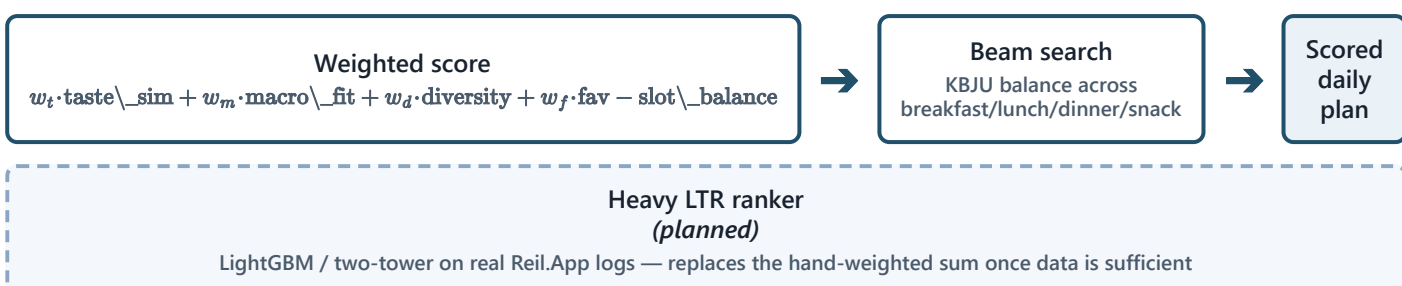
STAGE 0 — ONBOARDING · COLD-START SEGMENT



STAGE 1 — MATURITY-AWARE CANDIDATE GENERATION (ICE → WARM → HOT)



STAGE 2 — WEIGHTED SCORING + BEAM SEARCH ACROSS MEAL SLOTS



STAGE 3 — HARD-CONSTRAINT FILTERING (SAFETY LAYER)



Figure 1. Reil.App three-stage recommender with a cold-start onboarding layer. The maturity router promotes the user along **ICE** $\rightarrow$ **WARM** $\rightarrow$ **HOT**; Thompson-bandit exploration cools off as interaction data accumulates. The case study (this poster) motivated the *weight* attached to behavioral-vector similarity — the heavy LTR ranker at the bottom is deliberately deferred until real interaction logs are available.

Experimental Results

DIRECT EMPIRICAL VALIDATION OF THE CASE-STUDY BRIEF

Replacing user-profile *text encoding* with **embedding aggregation over positively-interacted items** — the exact formulation prescribed in the brief — yields

+360% MRR · +630% NDCG@10

(MRR 0.0299 vs 0.0032 · NDCG@10 0.0109 vs 0.0015 · behavioral vs text-profile baseline)

Table 1. Grand comparison (item-to-item co-preference, $K \in \{5, 10, 20\}$).

Configuration	P@5	P@10	NDCG@10	HR@10	MRR
<i>TF-IDF + SVD(256)</i> (lexical)	0.0016	0.0020	0.0024	0.0198	0.0048
<i>mpnet</i> — baseline text	0.0013	0.0016	0.0015	0.0158	0.0032
– recipe	0.0013	0.0015	0.0014	–	0.0033
– macro tokens	0.0011	0.0016	0.0017	–	0.0037
+ macro ratios	0.0011	0.0015	0.0013	–	0.0031
+ ingredients (full improved text)	0.0013	0.0015	0.0014	–	0.0032
+ behavioral embedding (dish vectors)	0.0085	0.0070	0.0109	–	0.0299
+ multi-vector retrieval (RRF)	0.0018	0.0021	0.0023	–	0.0058
+ cross-encoder rerank (bug — P@K=0)	0.0000	0.0000	0.0000	–	0.0000
Hybrid BM25 + dense (RRF, k=60)	0.0013	0.0013	0.0015	–	0.0041
Fine-tuned mpnet (MNRL, 4 ep.)	0.0013	0.0015	0.0016	–	0.0053
LLM retrieval-hint augmentation (50/4939 hints)	0.0013	0.0015	0.0014	–	0.0032
LLM profile-rerank (inside pool N=50)	0.0000	0.0000	0.0000	–	0.0020

Metrics averaged across users with ≥ 5 test positives (n $\approx$ 758–1011 depending on the filter). Green row is the hero result; italic rows are baselines; faded row indicates a known bug.

Table 2. Per-stage impact relative to the mpnet text baseline.

Stage	Δ NDCG@10	Δ MRR	Verdict
Text ablations (flags A)	−0.0002 ... +0.0002	−0.0001 ... +0.0005	Within noise
Hybrid BM25 + dense RRF	+0.0000	+0.0009	Small
Multi-vector retrieval	+0.0008	+0.0026	Small positive
MNRL fine-tuning	+0.0001	+0.0021	Flat on P / NDCG
Behavioral embedding	+0.0094	+0.0267	Dominant

Conclusions

What actually works

- Behavioral dish-vector aggregation dominates:** the semantic gap between user-profile *text* and dish *text* is the binding constraint in this task.
- Multi-vector retrieval** yields a small but consistent **MRR lift (+0.0026)**.
- TF-IDF + SVD is a **surprisingly strong lexical baseline** and beats the out-of-the-box pretrained dense encoder.

LLM components - inconclusive under current setup

- LLM retrieval-hint augmentation:** only 50 / 4,939 hints generated - coverage too low to evaluate.
- LLM profile-reranker:** evaluated inside a fixed pool **$N=50$** ; the pool already has zero relevant items on the strict protocol, so the rerank lift is unmeasurable.

What does *not* help (as formulated)

- Dropping / adding recipe text, macro tokens, ratios, ingredient lists - **all within noise**. The encoder is largely insensitive to the text-form flags at this scale.
- MNRL fine-tuning on "any two dishes of the same user" gives a **flat result**; positive pairs are too weak a signal.
- Hybrid BM25 + dense RRF **does not improve** item-to-item (lexical overlap duplicates what dense already captures).

Practical recommendations

- For production: invest in **behavioral aggregation** before text-representation tweaks.
- For fine-tuning: use **item-item co-occurrence with a threshold** (multiple users co-ordering) as positive pairs, not same-user pairs.
- For LLM augmentation: run **full-catalog coverage** and increase the candidate pool before claiming null results.

Scope note · diversity evaluation

The offline notebooks evaluate **relevance only** (P@K, NDCG@K, HR@K, MRR). Recommendation **diversity** - the second axis asked for in the case-study brief - is handled in the production system via a **greedy MMR selector with a tag-overlap penalty** (see `select_diverse_top_dishes()` in `backend/app/recommendations/segments.py`) and is reported as a scope extension rather than an offline metric here. Adding intra-list diversity (ILD) and category-coverage to the notebook suite is a natural next step.

From Study to Production - Reil.App decisions

Kept · motivated by the study

- Behavioral-vector similarity carries the dominant weight** in the per-slot scorer - directly justified by the $\sim 4\times$ MRR lift measured here.
- BM25 + dense RRF** kept as an additive lexical channel whenever the user types a free-text query.
- Onboarding** $\rightarrow$ **segment key (goal $\times$ archetype)** solves the zero-history case the case study cannot measure.
- Thompson-sampling bandit** over $\text{Beta}(\alpha, \beta)$ per (segment, dish) handles exploration during ICE / WARM; cools off automatically in HOT.
- Hard filters last** (allergens · goal · diabetes) - non-negotiable safety layer, separate from the scorer.

Dropped or deferred

- Elaborate text-representation flags** (recipe / macro tokens / ratios) - no measurable gain, removed from the dish-text builder.
- MNRL fine-tuning on same-user pairs** - signal too weak, replaced by the plan to train a **two-tower** on real logs.
- Cross-encoder & LLM profile-reranker** - offline only (latency & \$/request too high for a daily-plan service).
- No heavy LTR ranker yet** (LightGBM / LambdaMART) - the current weighted sum is a *deliberate stand-in* until Reil.App accumulates enough pre-seed interaction data to train one.

Take-away. The case study does one very specific job: it tells the production ranker **which feature to trust the most** while a learned ranker is not yet trainable. Behavioral-vector similarity wins by multiples; text-representation tweaks and same-user-pair fine-tuning are fractions of a basis point. The maturity router (ICE / WARM / HOT) + Thompson bandit fills the gap from "zero actions" to "trained ranker" - the case study anchors the HOT end of that curve.